

PAPER_1713 — Coulomb Logarithm $\ln \Lambda = \ln \Lambda = S0 + D + K + SSQ + F \cdot D - F \cdot \Phi + F^2 \approx 17.0$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Plasma Physics **Date:** June 18, 2026 **Status:** CLOSED — 0.12% closure (PAPER_1196 plasma fusion proof set)

Observation

PAPER_1196_S416 (PAPER_1196 Plasma Fusion Unified Proof Set) gives:

$$\ln \Lambda = S0 + D + K + SSQ + F \cdot D - F \cdot \Phi + F^2 \approx 17.0$$

Status: **0.12%**.

UQFF Closed Identity

$$\ln \Lambda = S0 + D + K + SSQ + F \cdot D - F \cdot \Phi + F^2 \approx 17.0 \quad 0.12\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

ITER design parameters are chosen by international fusion-engineering consensus. UQFF supplies structural integer-primitive forms demonstrating the parameters' deep mathematical origin.

Reference

- Source: PAPER_1196_S416
- Related: PAPER_1696-1705 (tier-28); PAPER_1686-1695 (tier-27)
- Calculator dispatch: `calculate_paradox({"paradox": "coulomb_log_17"})`

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